SMOOTHNESS OF A GROUPOID IN FUNCTORS IS EQUIVALENT TO SMOOTHNESS OF ITS SOURCE MAP AND OF ITS QUOTIENT MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-defos.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the comparison of the source and target maps through the inverse morphism of the groupoid, the explicit unwinding of the two smoothness conditions, and the translation between them using the inverse morphism.

Editorial note. The label lemma-smooth-quotient-morphism names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source formal-defos.tex –label lemma-smooth-quotient-morphism –start 6656 –end 6699 –local-id 28022 returned RESULT=OK (L28022 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-formal-defos/candidates-halfA.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Inverse morphism of the groupoid and the identity section. Same chapter, the defining data of a groupoid in functors, in particular the inverse morphism i : R -> R and the identity e : U -> R. Supplies the identities t = s composed with i and the elements used to translate between the two lifting conditions.
- Description of morphisms of the quotient groupoid. Same chapter, the remark describing the objects and morphisms of the quotient [U/R] as pairs with the compatibility conditions. Supplies the translation of a morphism (f, a) of the quotient into the data of ring maps and elements used in the unwinding.

Source reference. Original reference 06GN: sources/stacks-formal-defos/pinned-original-source.tex, source lines 1432–1432. Prerequisite pointer invoked at proof line(s) 6690; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 06GK: sources/stacks-formal-defos/pinned-original-source.tex, source lines 1381–1381. Prerequisite pointer invoked at proof

Date: stacks-project:formal-defos.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

line(s) 6689; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let (U, R, s, t, c) be a groupoid in functors on $\mathcal{C}_\Lambda$. The following are equivalent:*

1. *The groupoid in functors (U, R, s, t, c) is smooth.*
2. *The morphism $s : R \to U$ is smooth.*
3. *The morphism $t : R \to U$ is smooth.*
4. *The quotient morphism $U \to [U/R]$ is smooth.*

Proof. Statement (2) is equivalent to (3) since the inverse $i : R \to R$ of (U, R, s, t, c) is an isomorphism and $t = s \circ i$. By definition (1) is equivalent to (2) and (3) together, hence it is equivalent to either of them individually.

Finally we prove (2) is equivalent to (4). Unwinding the definitions:

(2) Smoothness of $s : R \to U$ amounts to the following condition: If $f : B \to A$ is a surjective ring map in $\mathcal{C}_\Lambda$, $a \in R(A)$, and $y \in U(B)$ such that $s(a) = U(f)(y)$, then there exists $a' \in R(B)$ such that $R(f)(a') = a$ and $s(a') = y$.

(4) Smoothness of $U \to [U/R]$ amounts to the following condition: If $f : B \to A$ is a surjective ring map in $\mathcal{C}_\Lambda$ and $(f, a) : (B, y) \to (A, x)$ is a morphism of $[U/R]$, then there exists $x' \in U(B)$ and $b \in R(B)$ with $s(b) = x'$, $t(b) = y$ such that $c(a, R(f)(b)) = e(x)$. Here $e : U \to R$ denotes the identity and the notation (f, a) is as in Remarks 06GK (06GN); in particular $a \in R(A)$ with $s(a) = U(f)(y)$ and $t(a) = x$.

If (4) holds and f, a, y as in (2) are given, let $x = t(a)$ so that we have a morphism $(f, a) : (B, y) \to (A, x)$. Then (4) produces x' and b, and $a' = i(b)$ satisfies the requirements of (2). Conversely, assume (2) holds and let $(f, a) : (B, y) \to (A, x)$ as in (4) be given. Then (2) produces $a' \in R(B)$, and $x' = t(a')$ and $b = i(a')$ satisfy the requirements of (4). □